

Machine-Verified Results in Quasi-Amicable (Betrothed) Number Theory

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1 Machine-Verified Results in Quasi-Amicable (Betrothed) Number Theory

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· Author: BCC / Brockian Mathematics program Verification: local lake build (Lean toolchain pinned at leanprover/lean4:v4.32.0 + Mathlib) and the AXLE cloud kernel (Lean 4.32.0). No `sorry`, `admit`, or added `axiom` appears in any file cited below.

1.1 1. What betrothed numbers are

Write $s(n)$ for the **aliquot sum** — the sum of the proper divisors of n . A pair (m, n) of distinct positive integers is **betrothed** (also called **quasi-amicable** or **reduced amicable**) when each is *one more* than the aliquot sum of the other:

$$s(m) = n + 1 \text{ and } s(n) = m + 1.$$

They are the “off-by-one” analogue of amicable pairs ($s(m) = n, s(n) = m$). The smallest is (48, 75); then (140, 195), (1050, 1925), (1575, 1648), (5775, 6128), This is OEIS **A005276**. Two questions are **classically open**: whether there are **infinitely many** betrothed pairs, and whether any pair has **both members of the same parity** (every known pair is one even and one odd number).

This note reports what we have **machine-verified**, and — importantly — separates it into three registers: what is *classical and not ours*, what is a *new formalization*, and what is a *genuinely new (small) theorem*. It also states plainly what remains open.

1.2 2. The results, by register

1.2.1 2a. CLASSICAL — not our discovery

The **two-cycle / quasi-cycle viewpoint** of betrothed numbers, and the structural restrictions on their form (e.g. the same-parity case forcing a square or twice-a-square), are due to **Hagis & Lord**, “Quasi-Amicable Numbers,” *Math. Comp.* **31** (1977), 608–611. That betrothed numbers have **density zero** is **Pollack**, “Quasi-Amicable Numbers are Rare,” *J. Integer Seq.* **14** (2011). We claim none of this; we build on it and cite it.

1.2.2 2b. NEW FORMALIZATION — the exact Lean statements in this repository

In `Brockian/BetrothedNumbers/Dynamics.lean` (integrated, builds locally on `Brockian/BetrothedNumbers.lean`) we give an exact Lean rendering of the classical two-cycle picture and an exact abundance **balance law** for the standard $2^k \cdot p$ construction family.

Define the **partner map** `partner n := s(n) - 1`. Betrothed pairs are exactly its nontrivial two-cycles:

```
def partner (n :  $\mathbb{N}$ ) :  $\mathbb{N}$  := aliquot n - 1

theorem betrothed_iff_twoCycle {m n :  $\mathbb{N}$ } :
  Betrothed m n
  ↔ m  $\neq$  n - 1  $\wedge$  aliquot m - 1  $\leq$  aliquot n - 1  $\wedge$  partner m = n  $\wedge$  partner n = m

theorem partner_involutive_on_pair {m n :  $\mathbb{N}$ } (h : Betrothed m n) :
  partner (partner m) = m

theorem partner_ne_self_on_pair {m n :  $\mathbb{N}$ } (h : Betrothed m n) : partner m  $\neq$  m
```

For the construction family $2^k \cdot p$, the forced odd partner is `thabitPartner k p = (2k - 1)(p + 2)`. Under an explicit **σ -criterion hypothesis** (that σ of that partner equals $(2^{k+1} - 1)(p + 1)$), the sign of $(p + 3) - 2^{k+1}$ fixes the abundance class of the partner exactly:

```
theorem thabit_balance_identity {k p :  $\mathbb{N}$ } (hk : 1  $\leq$  k)
  (hcriterion :  $\sigma$  1 (thabitPartner k p) = (2k+1 - 1) * (p + 1)) :
   $\sigma$  1 (thabitPartner k p) + 2k+1 = 2 * thabitPartner k p + (p + 3)

-- and the three phase-boundary corollaries, each CONDITIONAL on the same hcriterion:
theorem thabit_deficient_iff ... :  $\sigma$  1 (...) < 2 * (...)      p + 3 < 2k+1
theorem thabit_perfect_iff   ... :  $\sigma$  1 (...) = 2 * (...)      p + 3 = 2k+1
theorem thabit_abundant_iff  ... : 2 * (...) <  $\sigma$  1 (...)      2k+1 < p + 3
```

Honesty note: the balance law and its deficient/perfect/abundant corollaries are **conditional** on the σ -criterion — it is an explicit hypothesis in each signature, *not* an unproven axiom, and this note does not claim it holds unconditionally. A self-contained twin of these statements, corrected to the repo’s actual `aliquot/Betrothed` definitions, is AXLE-verified in `aristotle/betrothed_faithful.lean` (Lean 4.32.0). The exact Lean two-cycle reformulation and the phase-boundary balance law appear to be **new to this repository**. A modern Lean amicable-numbers library — **Chen, Tang, Zhan**, “Formalization of Amicable Numbers

Theory,” arXiv:2601.07444 (2026) — includes a betrothed layer that, per their paper, is a *definition plus the two smallest examples*; our formalization is materially more, **pending a direct source-level comparison** (we have not diffed their Lean sources, so we do not assert superiority as fact).

1.2.3 2c. NEW DEDUCTION — a genuinely original small theorem

The one result here that is, to our knowledge, **new mathematics** (not merely a formalization of something known) is an **unconditional obstruction**, machine-verified in `aristotle/best_proofs/Brockian_BetrothedNumbers_no_pair_of_mersenne_and_shifted_prime.lean`:

```
/-- Negative obstruction. Let  $k \geq 2$  and let  $p$  be an odd prime such that both
 $2^k - 1$  and  $p + 2$  are prime. Then  $2^k * p$  has no betrothed partner. -/
theorem no_pair_of_mersenne_and_shifted_prime {k p :  $\mathbb{N}$ }
  (hk :  $2 \leq k$ ) (hp : p.Prime) (hodd : Odd p)
  (hq : Nat.Prime (2 ^ k - 1)) (hr : Nat.Prime (p + 2)) :
  ¬ ∃ m :  $\mathbb{N}$ , IsBetrothedPair m (2 ^ k * p)
```

Here $\text{IsBetrothedPair } m \ n := 0 < m \ 0 < n \ \sigma \ m = m + n + 1 \ \sigma \ n = m + n + 1$ (distinctness is *not* assumed, which only strengthens the non-existence claim). The theorem is **unconditional**: its hypotheses are exactly $k \geq 2$, p an odd prime, and the two auxiliary primality conditions ($2^k - 1$ a Mersenne prime, $p + 2$ a prime). There is no σ -criterion hypothesis and no `sorry/axiom` in the signature or proof — it is a complete kernel-checked deduction. The proof route: a **unique-partner lemma** (`betrothed_partner_eq`) forces any partner of $2^k \cdot p$ to equal $(2^k - 1)(p + 2)$; then when both factors are prime, the sum-of-divisors identity $(q+1)(p+3) = q(p+2) + (q+1)p + 1$ collapses to $q \cdot p = q + 2$, which is impossible for $q \geq 3$, $p \geq 3$. A degenerate square sub-case ($q = p+2$) is ruled out separately.

Plainly: this is a **small but genuine original contribution** to quasi-amicable number theory — a clean sufficient condition under which a member of the $2^k \cdot p$ family provably has no betrothed partner.

1.2.4 2d. Computational evidence (COMPUTATION register — evidence, not proof)

Independently reproduced from scratch (`aristotle/betrothed_verification_report.json`): a falsifier over **996,888** parameter pairs ($1 \leq k \leq 24$, odd prime $p \leq 500000$) found **exactly three** betrothed solutions in the $2^k \cdot p$ family — (48, 75) deficient, (1648, 1575) abundant, (6128, 5775) abundant — and among the **31,955** cases where *both* auxiliary numbers $2^k - 1$ and $p + 2$ are prime, **zero** successes. That zero is the computational *evidence* consistent with the obstruction theorem above; it proves nothing beyond the range searched. A bonus empirical finding: the construction identity $B(2^k \cdot p) = (2^k - 1)(p + 2)$ held for **all** 996,888 pairs. Concrete pairs are also verified as standalone kernel theorems, e.g. `betrothed_48_75`, `betrothed_140_195`, `betrothed_1050_1925` (repo) and `betrothed_5775_6128` (`aristotle/best_proofs/..._5775_6128.lean`).

1.2.5 2e. NOT RESOLVED — open questions we did not touch

- **Infinitude of betrothed pairs** — OPEN. Recorded in the repo as an unproven `def BetrothedInfinitude`; we do not prove it.

- **Same-parity betrothed pair** — OPEN in both directions. Recorded as `def SameParityBetrothedExists`. We verify only that the two smallest known pairs have *opposite* parity; that is an illustration, not a resolution.
- The obstruction is a *sufficient* non-existence condition for one construction family; it says nothing about betrothed pairs outside $2^k \cdot p$, and nothing about density (Pollack already has density zero).

1.3 3. What is new vs. classical vs. open

Item	Register	Ours?	Verified how
Two-cycle / quasi-cycle viewpoint	Classical (Hagis–Lord 1977)	No	—
Same-parity $\Rightarrow$ square / twice-square restriction	Classical (Hagis–Lord 1977)	No	—
Density zero	Classical (Pollack 2011)	No	—
Exact Lean two-cycle reformulation (<code>betrothed_iff_twoCycle</code> , <code>partner</code> <code>involutivity/ne-self</code>)	New formalization	Yes (repo)	<code>lake build</code> , Lean 4.32.0
$2^k \cdot p$ abundance balance law + defi- cient/perfect/abundant phase boundary	New formalization , conditional on σ -criterion	Yes (repo)	<code>lake build</code> + AXLE, Lean 4.32.0
Mersenne / shifted-prime obstruction (<code>no_pair_of_mersenne_</code> <code>and_conditional_prime</code>)	New deduction (original small theorem), and conditional prime	Yes	AXLE kernel, Lean 4.32.0
996,888-pair falsifier; 0/31,955 both-prime successes	Computation (evidence)	Yes (reproduced)	independent script
Concrete pair witnesses (48/75, ..., 5775/6128)	Computation / kernel	Yes	<code>decide</code> / <code>norm_num</code> , kernel
Infinitude of betrothed pairs	OPEN	—	not attempted
Same-parity existence	OPEN	—	not attempted

1.4 4. Provenance and verification method

- **Local repo module:** Brockian/BetrothedNumbers.lean (definitions aliquot, Betrothed) and Brockian/BetrothedNumbers/Dynamics.lean (two-cycle reformulation + balance law). Builds with lake build; toolchain pinned at leanprover/lean4:v4.32.0 + Mathlib.
- **AXLE-verified self-contained twin:** aristotle/betrothed_faithful.lean (Lean 4.32.0), rebuilt against the repo’s real definitions after the originally delivered BetrothedDynamics.lean was found to reference a nonexistent IsBetrothedPair predicate.
- **Unconditional obstruction:** aristotle/best_proofs/Brockian_BetrothedNumbers_no_pair_of_mers produced via the Aristotle prover and kernel-checked. Confirmed no unproven hypotheses in the signature and no sorry/axiom in the proof.
- **Computation:** aristotle/betrothed_verification_report.json, independently reproduced from scratch.
- **Literature boundary:** Hagi & Lord (1977); Pollack (2011); Chen–Tang–Zhan, arXiv:2601.07444 (2026); OEIS A005276.

Open review item for Chris: the prompt referenced Lean 4.33.0-rc2 for the local build, but the repo’s lean-toolchain currently pins v4.32.0 and the module headers record 4.32.0 — this note states 4.32.0 to match the files as they stand. Worth confirming before any external write-up. The comparison to the 2026 Chen–Tang–Zhan betrothed layer is stated as “materially more, pending direct source comparison” and should be checked against their actual Lean sources before any public claim.